Topic 6

Mental Workload and Automation

6.1. Work physiology

6.2. Cognitive load, stress, and fatigue

6.3. Disruption and desynchronization

6.4. Classes of automation

6.5. Human-centred automation

6.6. Automation-based complex systems

Mental Workload

The **cognitive demand** placed on an individual while performing a task.

Importance

1. **Balancing workload** to prevent fatigue and errors.

2. Ensuring **effective** human-machine collaboration.

6.1

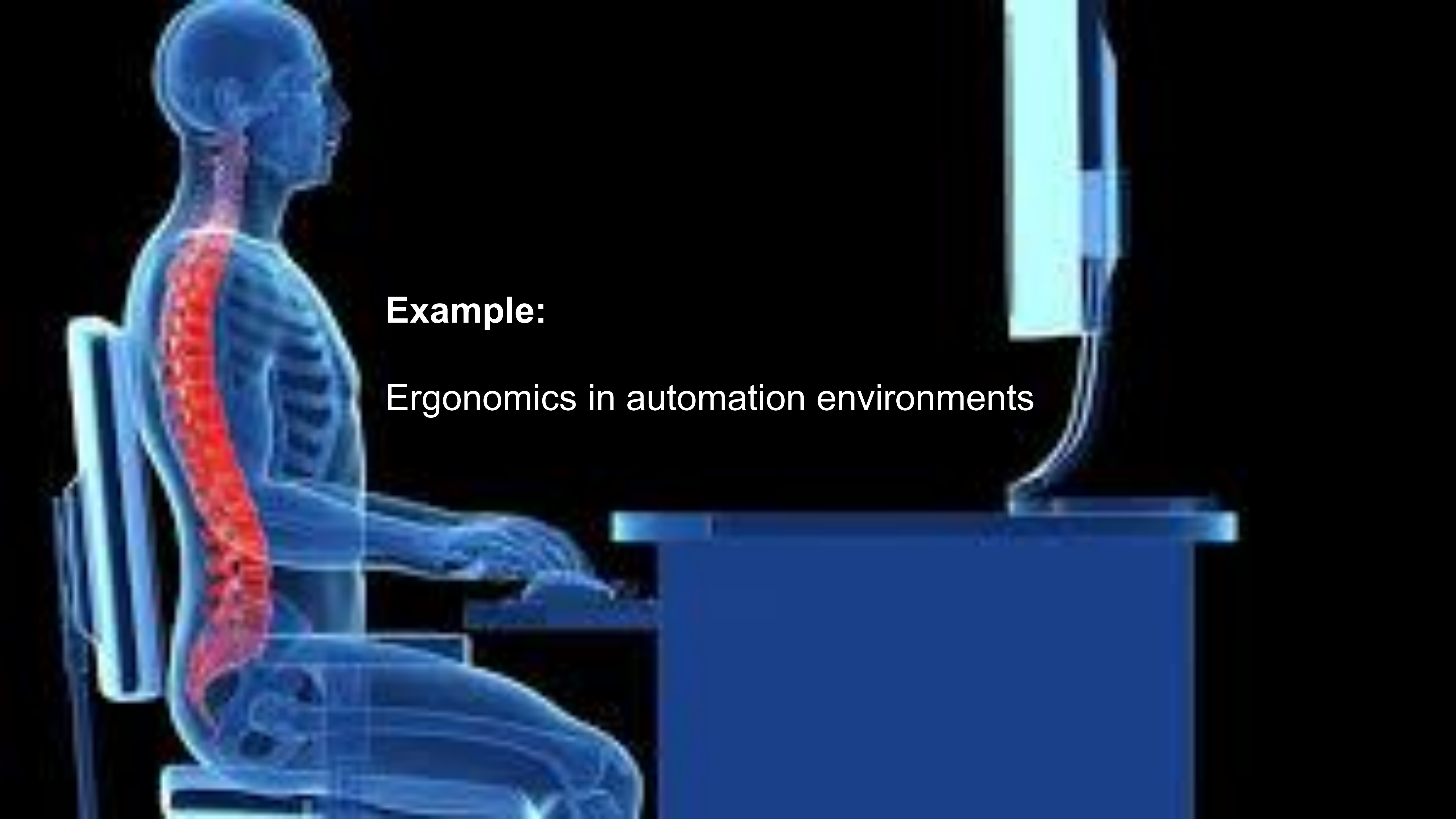
Work Physiology

A study of **physical** and **physiological** demands of work on the human body.

Examines **how the human body responds** to physical and mental tasks.

Studies the interactions between physical demands, task performance, and the physiological systems of the body, such as the cardiovascular, respiratory, and muscular systems.

- Physical vs. mental workload
- Impact on posture, energy consumption, and stress levels



Example:

Ergonomics in automation environments

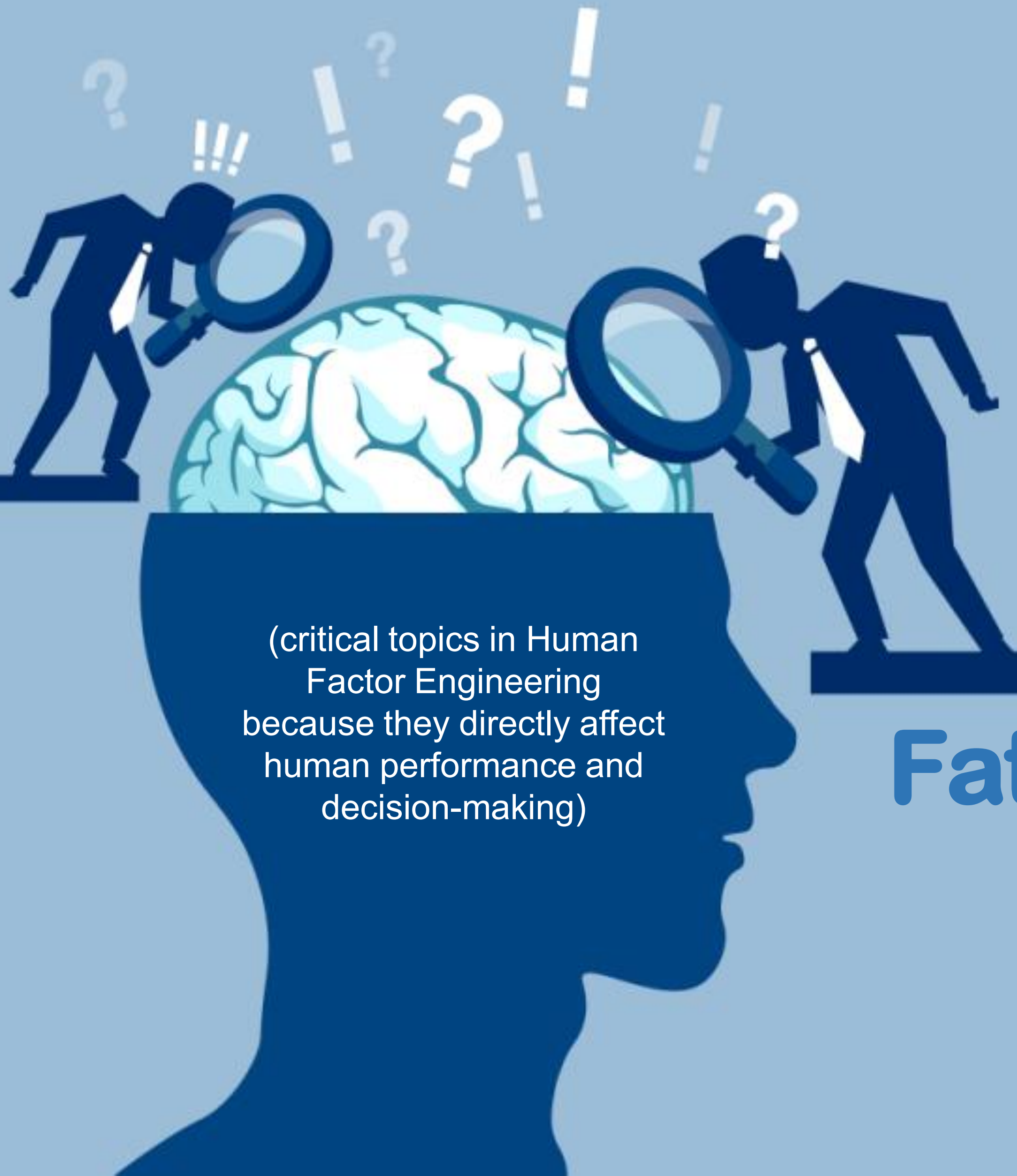
Understanding work physiology, tasks and environments can be optimized to:

1. Enhance performance
2. Reduce injuries
3. Ensure overall well-being

This is crucial in Human Factor Engineering, where designing **systems that align with human physical capabilities** is a priority

6.2

Cognitive load



Stress

(critical topics in Human Factor Engineering because they directly affect human performance and decision-making)

Fatigue

Cognitive load

Refers to **the amount of mental effort** being used in the working memory during task performance.

It can vary depending on task complexity and the individual's capacity.

Types of Cognitive Load:

- **Intrinsic Load:** Linked to the complexity of the task itself.
- **Extraneous Load:** Caused by how information is presented, such as poor design or unclear instructions.
- **Germane Load:** Associated with the process of learning and transferring knowledge.

Relevance in Human Factors:

- High cognitive load can reduce task efficiency and lead to errors
- Tools like checklists or simplified interfaces can help manage cognitive load

Stress

A physical or emotional response to external or internal pressures

The feeling of **anxiety** and **discomfort** that the user of a computer system suffers when the computer system does not work or does user’s expectations.

Effects on Performance:

1. Low levels of stress can enhance focus and performance (eustress).
2. High stress levels can impair cognitive functions, reduce attention span, and lead to burnout.

Mitigation Strategies:

1. Training for stress management
2. Ergonomic design
3. Supportive work environments.

Fatigue

A state of **mental or physical exhaustion** that reduces the ability to perform tasks effectively

Causes of Fatigue:

1. Prolonged mental or physical activity
2. Insufficient rest
3. Monotonous tasks

Impact on Performance:

1. Fatigue decreases alertness
2. Impairs decision-making
3. Increases the likelihood of errors or accidents

Management in Work Systems:

1. Incorporating breaks
2. Workload adjustments
3. Designing tasks to reduce monotony

Integration in System Design

1. **Minimizing Cognitive Load**
   Use intuitive interfaces, provide clear instructions, and limit the amount of information presented at one time.

2. **Stress Reduction**
   Incorporate feedback mechanisms and error-tolerant systems that help users manage high-pressure scenarios.

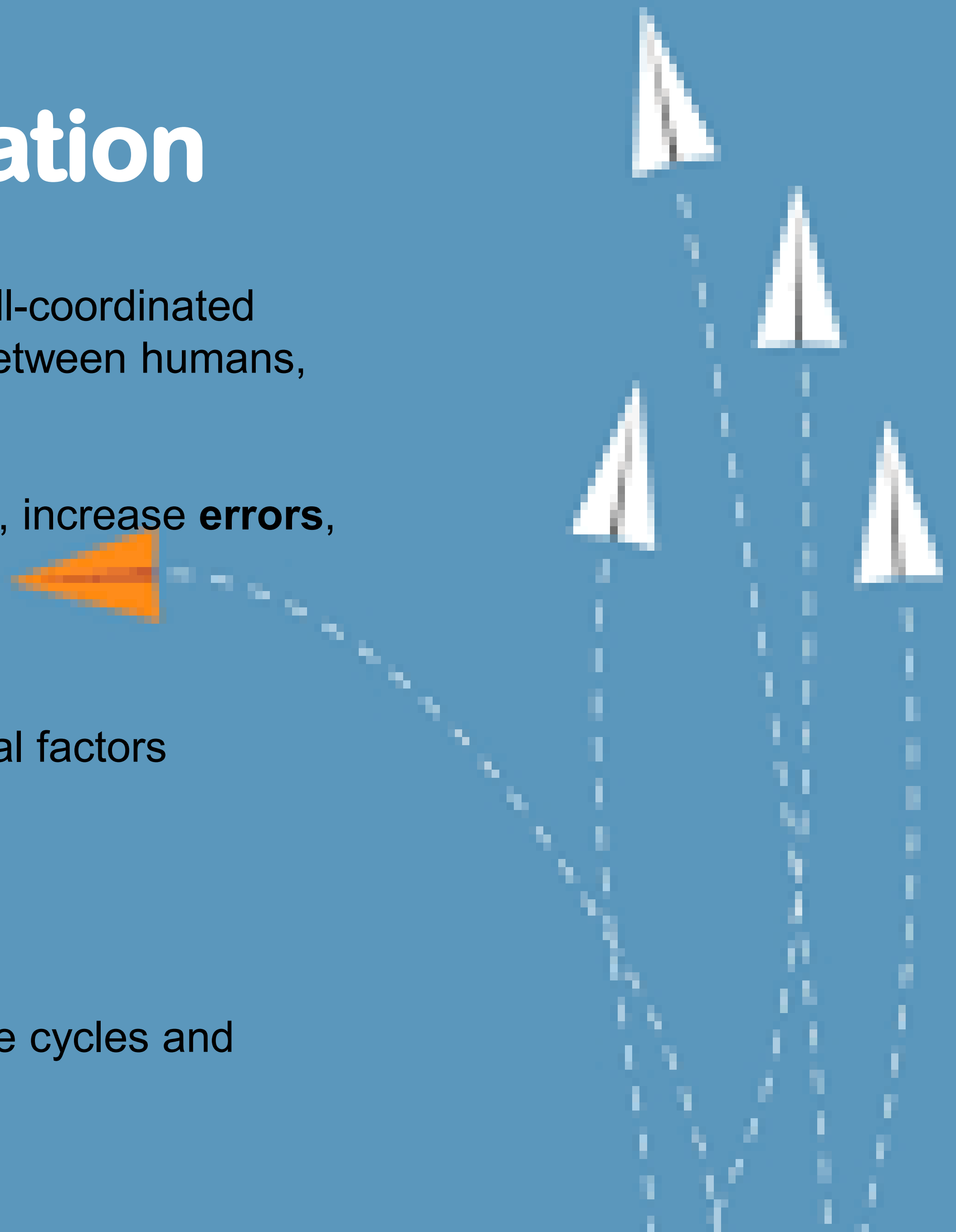
3. **Addressing Fatigue**
   Monitor workload and offer automation tools to assist in repetitive or demanding tasks.

6.3

Disruption & desynchronization

Breakdowns or **misalignments** in well-coordinated systems, often involving interactions between humans, technology, and the environment

Impact: **task performance**, **efficiency**, increase **errors**, **safety** in complex systems



Disruption

Interruptions in task flow due to external factors (e.g., alarms, system errors)

Desynchronization

Misalignment between human cognitive cycles and machine outputs

Disruption

Occurs when an unexpected event or change interferes with normal operations or workflows

1. interruptions during tasks
2. sudden changes in workload
3. external disturbances such as alarms or environmental changes

Source of Disruption

1. technology (system failures)
2. environment (power outages)
3. human factors (errors or miscommunications).

Impact

1. Delays
2. Errors
3. reduced efficiency

Mitigation

1. robust system design
2. user training
3. contingency planning

Desynchronization

the loss of synchronization between interdependent components in a system

Team Coordination
Teams working on interdependent tasks may face delays if their actions are not aligned

Human-Machine Interaction
automation in a system is poorly designed, it may operate out of sync with human actions, leading to confusion or errors

Biological Desynchronization
Misalignment between human circadian rhythms and work schedules (shift workers or jet-lagged individuals)

Relevance in Human Factors Engineering

Cognitive Load
High cognitive demands during disruptions can lead to errors or oversight

System Design
Designing systems that account for disruptions and maintain synchronization enhances resilience and user experience

Task Performance
Synchronization ensures smooth workflows and prevents bottlenecks

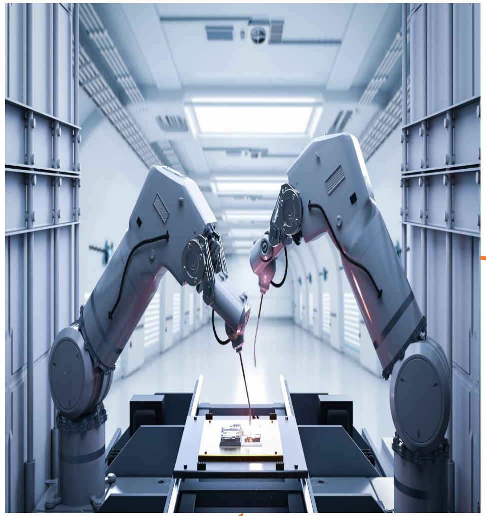
By understanding and mitigating disruption and desynchronization, designers can create systems that are more **robust**, **adaptive**, and **user-friendly**

6.4 Classes of automation

Automation levels based on the level of **involvement of humans and machines** in completing tasks.

The classifications are designed to measure how responsibility and control are shared between humans and automated systems

Sheridan & Verplank's Levels of Automation



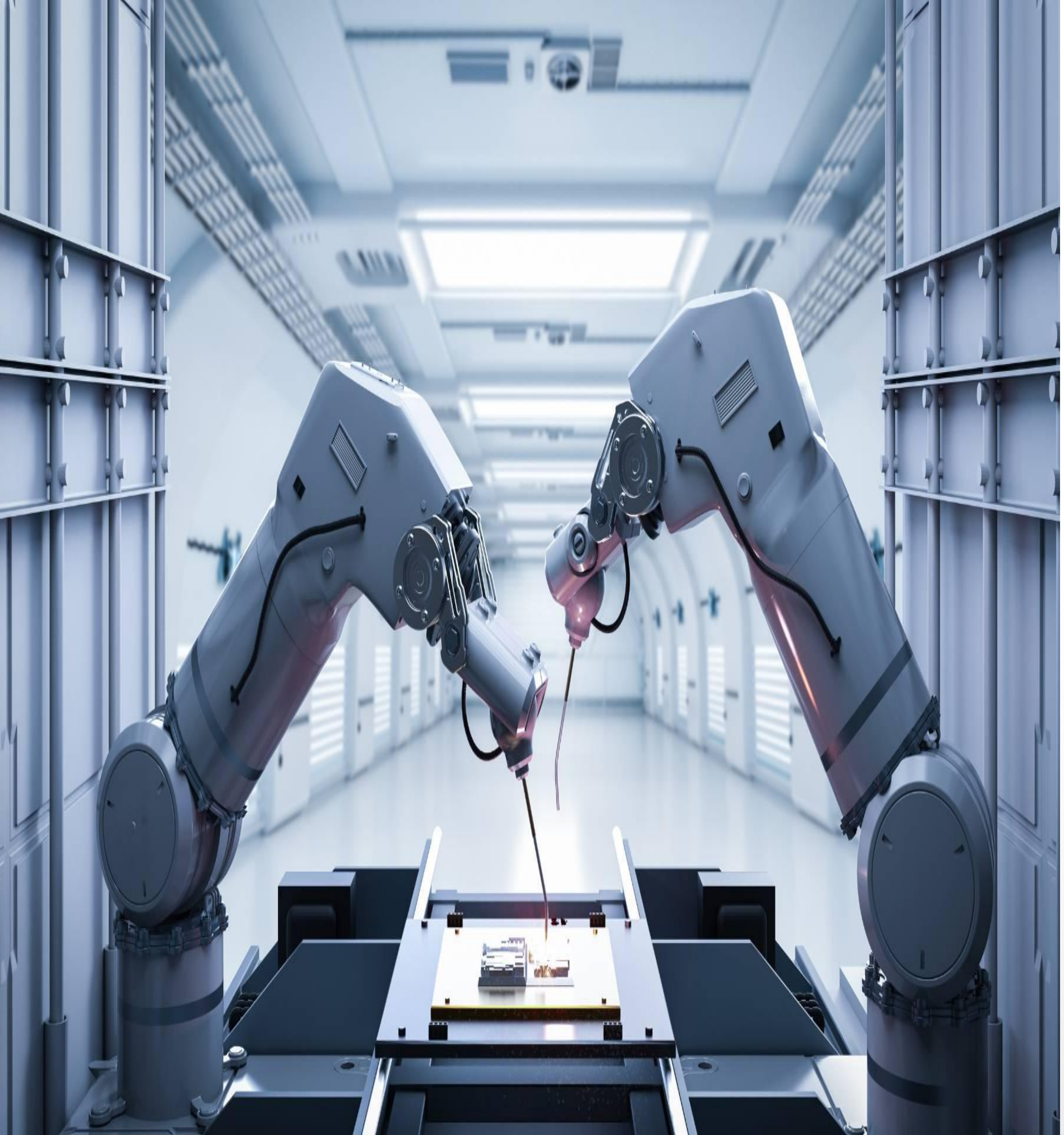
Manual Control
Humans perform all tasks with no automation

Assisted Control
Automation provides suggestions, but humans make final decisions (e.g., driver-assist features in cars).

Shared Control
Humans & machines **share control dynamically** based on context or workload

Supervised Automation
The system handles most tasks, but human operators intervene when necessary (e.g., autopilot in airplanes).

Full Automation
Machines independently execute tasks without human intervention (e.g., autonomous vehicles)



Classes of Automation Based on Functional Aspects

Information Acquisition and Analysis

Automation collects, processes, and interprets data to assist decision-making (e.g., diagnostic tools in healthcare).

Decision and Action Selection

Automation recommends or makes decisions based on predefined rules or AI algorithms (e.g., air traffic control decision aids).

Action Execution

Automation physically performs tasks (e.g., robotic arms in manufacturing).

Lower Levels of Automation

maintain high human control, risk of increased workload and errors under stress

Higher Levels of Automation

reduce human workload but may lead to **over-reliance** or loss of situational awareness

Challenges & Considerations

When designing systems with varying levels of automation:

Human-Centered Design

Ensure that automation **complements**, rather than replaces, human capabilities

Adaptability

The automation level should adapt based on context and operator expertise

Error Mitigation

Higher automation must include **fail-safe mechanisms** to avoid catastrophic consequences if the system fails

6.5 Human-centred automation

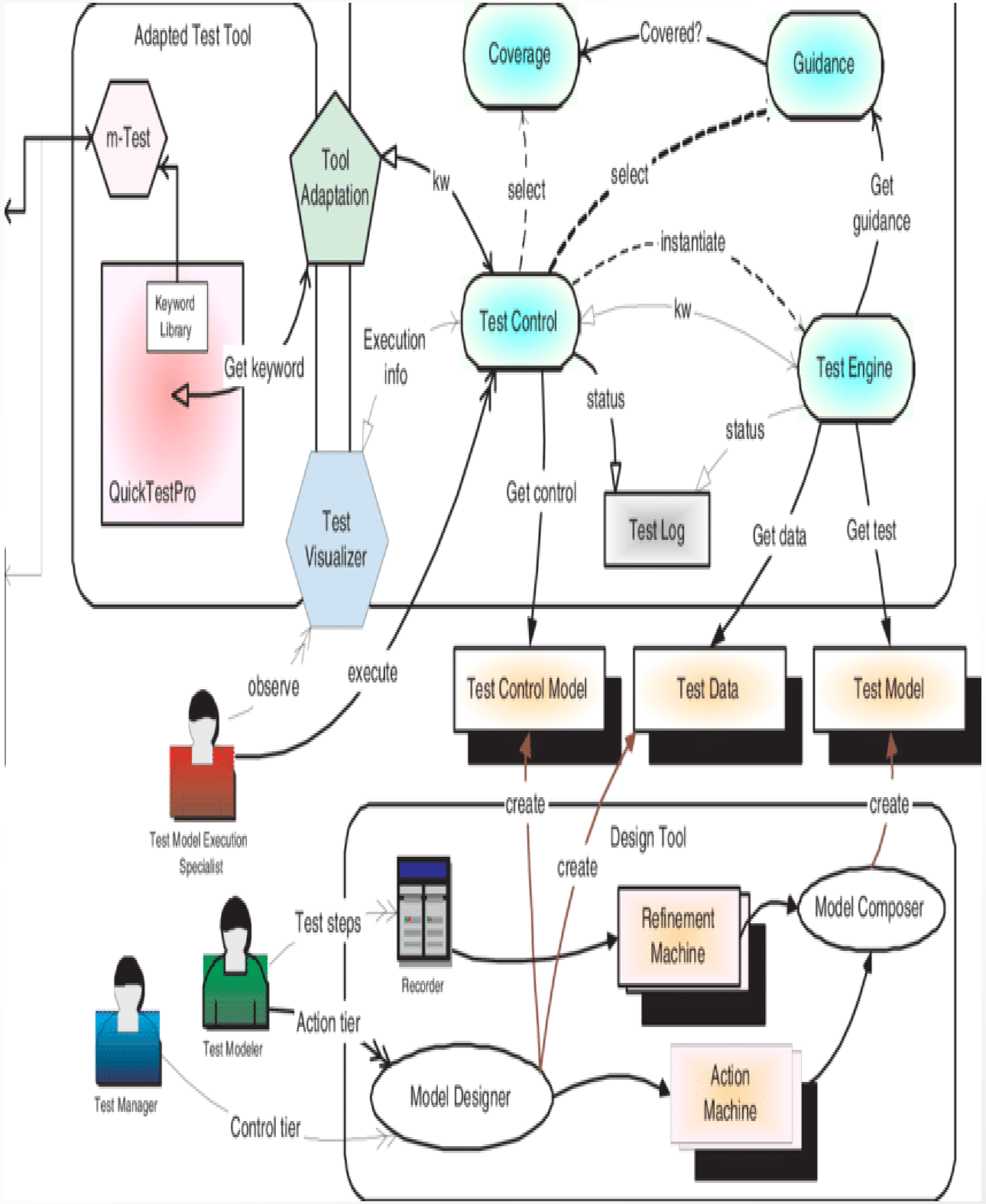
Designing systems to prioritize human needs and limitations

Principles:

1. High User involvement in system design.
2. Clear feedback and control mechanisms.

Benefits:

1. Enhanced trust and adoption.
2. Reduced error rates.



6.6 Automation-based complex systems

Systems involving **multiple automated processes** and human oversight

Examples

- Air traffic control.
- Power grid monitoring.

Benefits

1. Reduces human workload, stress, and error rates.
2. Enables real-time data analysis and decision-making.
3. Improves safety and operational efficiency.

Challenges

1. Cognitive overload due to system complexity.
2. Communication between automated subsystems.
3. System reliability and resilience against failures.
4. Balancing automation with human control to avoid over-reliance.
5. Addressing ethical and social implications, such as job displacement.

Solutions

1. Simplified interfaces
2. Training for operators on system behavior